

Project Overview

Fraud Detection Analytics Case Study

Background

Financial fraud poses significant challenges to banks, payment providers, and financial institutions by increasing operational costs, financial losses, and customer risk. Detecting suspicious transactions quickly is essential for reducing fraud exposure and improving decision-making.

This project presents an end-to-end fraud analytics workflow using SQL, Python, Power BI, and Tableau to analyze transaction data, identify fraud patterns, and develop interactive dashboards that support business monitoring and reporting.

Business Problem

Financial institutions process thousands of transactions every day, making manual fraud detection impractical. Analysts require automated methods to identify suspicious behavior, monitor fraud trends, and prioritize high-risk transactions for investigation.

The objective of this project is to demonstrate how data analytics can transform raw transaction data into actionable business insights for fraud detection.

Objectives

- Analyze financial transaction data to identify potential fraud.
- Detect unusual customer and transaction behavior.
- Evaluate fraud trends across regions and time periods.
- Build interactive dashboards for fraud monitoring.
- Generate business recommendations based on analytical findings.

Analytics Workflow

- 1 Data collection and inspection
- 2 Data cleaning and preprocessing
- 3 Exploratory Data Analysis (EDA)
- 4 SQL-based fraud analysis
- 5 KPI calculation
- 6 Dashboard development in Power BI and Tableau
- 7 Business insight generation

Business Impact

The analytical approach demonstrated in this project can help organizations:

- Monitor fraud activity more effectively.
- Identify high-risk customers and transactions.

- Support fraud investigation teams with actionable insights.
- Improve operational decision-making using interactive dashboards.
- Enable data-driven fraud prevention strategies.

Tools Used

- Python
- Pandas
- NumPy
- SQL
- Power BI
- Tableau
- Microsoft Excel